

Module 5: Advanced Pattern Mining & Applications

Multilevel Rules, Antimonotonic Constraints, Colossal Patterns

Module V: Advanced Pattern Mining & Applications — Topics Covered

1. Multilevel & Multidimensional Association Rules

2. Quantitative Association Rule Mining

3. Constraint-Based Mining (Antimonotonic, Monotonic, Succinct)

4. Mining Colossal & Compressed Patterns

5. Real-World Applications (Retail Basket, Fraud Detection)

1. Constraint Properties in Frequent Itemset Mining

Constraint Type	Formal Property Definition	Pushing Strategy in Mining
Antimonotonic	If itemset S violates constraint C , all supersets $S' \supset S$ also violate C . (e.g., $\text{sum}(S.\text{price}) \leq 100$).	Prunes itemset search branch immediately (like Apriori support).
Monotonic	If itemset S satisfies constraint C , all supersets $S' \supset S$ also satisfy C . (e.g., $\text{sum}(S.\text{price}) \geq 500$).	Once satisfied, all descendant supersets need not be tested against C .